

Liver Function Tests (LFTs)

Prepared by

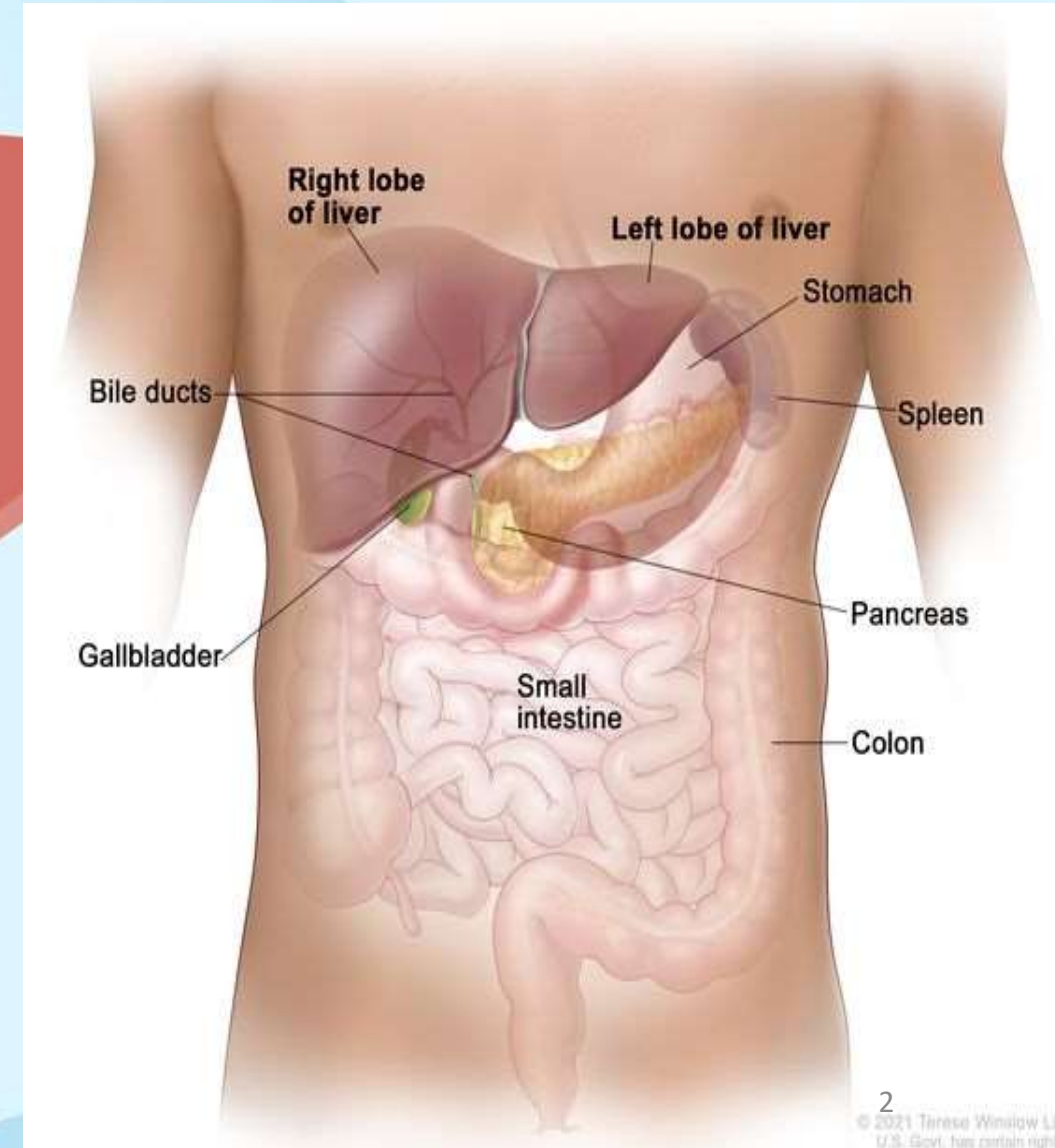
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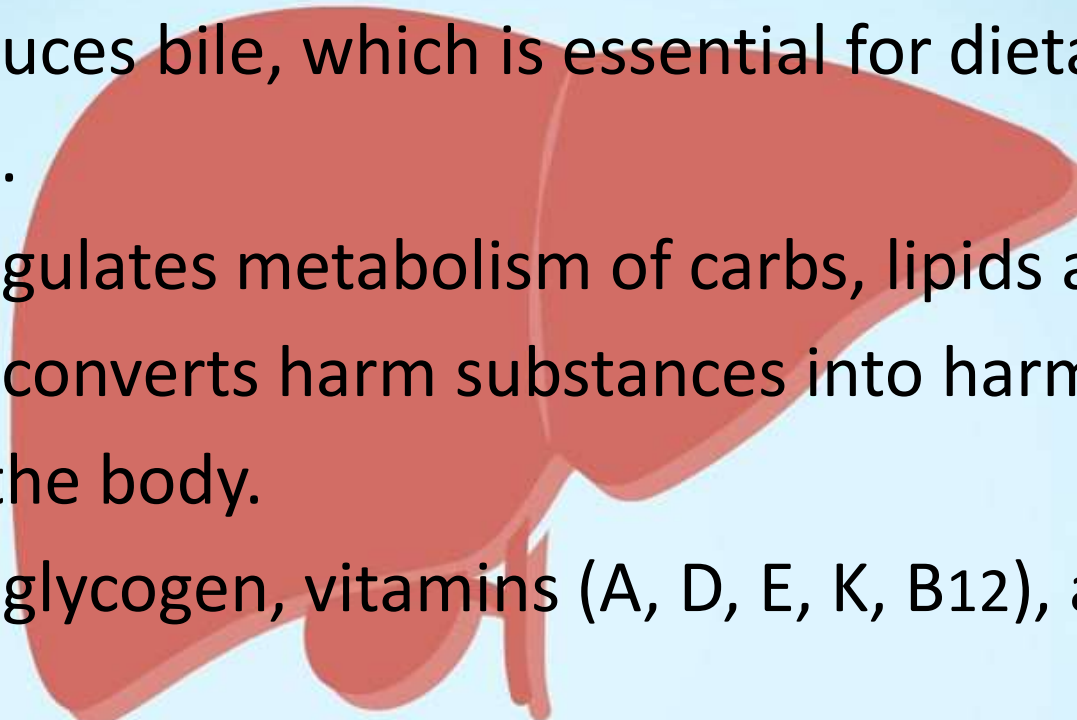
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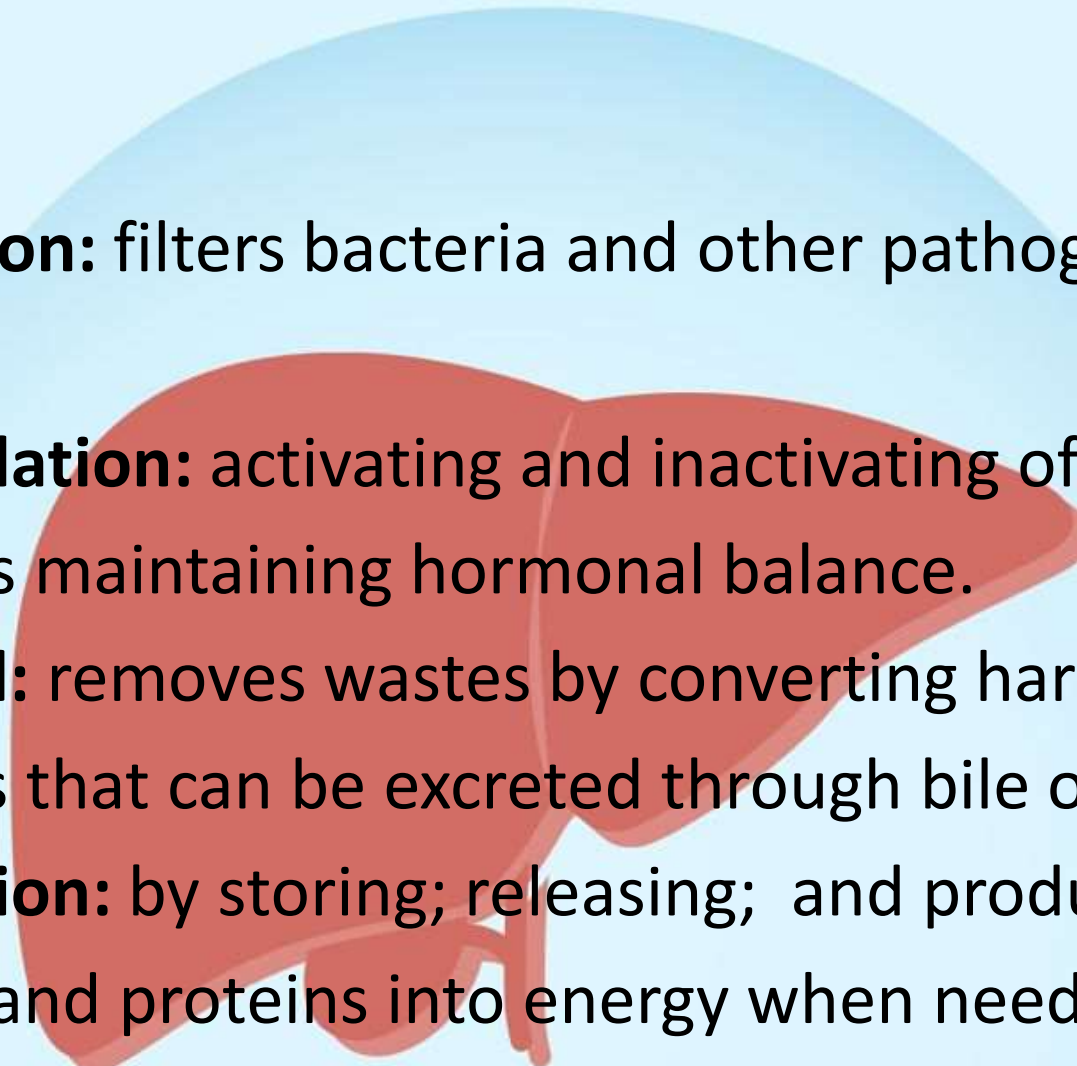
Introduction:

-  The liver is a large organ found in the upper right quadrant of the abdomen,
-  shaped like a cone, and it has a dark reddish-brown color that weighs about 1.5 Kg.
-  It's an essential organ of the body that performs over 500 vital functions.



Main Functions of the Liver:

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1. **Digestion:** produces bile, which is essential for dietary fat digestion and absorption.
 2. **Metabolism:** regulates metabolism of carbs, lipids and proteins.
 3. **Detoxification:** converts harm substances into harmless that can be excreted from the body.
 4. **Storage:** stores glycogen, vitamins (A, D, E, K, B12), and minerals (iron and copper).
 5. **Synthesis:** produces essential proteins and clotting factors, carbohydrates, lipids, bile salts, and hormones.

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- 6. Immune Function:** filters bacteria and other pathogens from the bloodstream.
 - 7. Hormone Regulation:** activating and inactivating of certain hormones, thus maintaining hormonal balance.
 - 8. Waste Removal:** removes wastes by converting harmful substances into substances that can be excreted through bile or urine.
 - 9. Energy Regulation:** by storing; releasing; and producing glucose, converting fats and proteins into energy when needed.

What are liver function tests?



Liver function tests (LFTs) or liver chemistries, are a panel of blood tests that measure **enzymes**, **proteins**, and **other substances** to assess liver health and function.



They help diagnose liver diseases, monitor their severity, and check for liver damage caused by different conditions.

The most common liver tests include:

1- Liver enzymes test: include alanine aminotransferase (ALT), aspartate aminotransferase (AST), alkaline phosphatase (ALP), and gamma glutamyl transferase (GGT). These are elevated when there's **liver injury**.

i- ALT:



Alanine aminotransferase is more specific for liver injury than AST.



Found mainly in the liver.



ALT transfers the amino group from alanine to α -ketoglutarate



Normal range: (5-45) IU/L



Very high levels are most often caused by viral hepatitis, ischemic hepatitis, or injury from drugs or other chemicals.

ii- AST:

-  Aspartate aminotransferase is less liver specific than ALT.
-  Found mostly in the **liver**, but also in the **heart**, **kidneys**, **skeletal muscle**, and **brain**.
-  AST transfers the amino group from aspartate to α -ketoglutarate.
-  Normal range: **(5-45) IU/L**
-  Elevated levels can indicate liver damage or injury (hepatitis, cirrhosis), MI, muscular dystrophy, kidney disease.

Note:

AST > ALT (alcoholic liver disease)

ALT > AST (viral or non- alcoholic liver disease)

iii- ALP:

- 🦋 Alkaline phosphatase is a **non-specific** marker of **liver disease**.
- 🦋 Found mainly in the **liver, bones, bile ducts, and placenta**.
- 🦋 ALP hydrolyzes phosphate groups from various compounds at an alkaline pH.
- 🦋 Normal range: **(30-120) IU/L**
- 🦋 High levels of ALP may indicate **liver inflammation, blockage of the bile ducts, and bone diseases**.

IV- GGT:

- 🦋 γ -glutamyl transferase plays an important role in glutathione metabolism & recycling.
- 🦋 An enzyme that is mainly made in the **liver, kidney, and biliary tract**.
- 🦋 GGT transfers the γ -glutamyl group from glutathione to an amino acid.
- 🦋 Normal range: **(5-50)IU/L**
- 🦋 Elevated in **liver and bile duct disorders, chronic alcohol consumption**.

2- Total protein test: measures the total amount of protein in the blood, which includes **albumin** and **globulins**. These proteins are mainly made in the liver.

i- Albumin:

-  The most abundant protein synthesized by the **liver**.
-  Normal range: **(3.5 to 5) g/dL**
-  It's level **decrease** in all **chronic liver diseases** like (cirrhosis, hepatitis, or liver cancer).
-  High levels of albumin **rarely** due to liver problems.

ii- Globulin:

-  The globulins are a family of globular proteins including (Gamma globulins known as immunoglobulins, Alpha and beta globulins, and blood clotting globulins).
-  Made by the **liver** and **immune system**.
-  Normal range: **(2.5 to 3.5g/dL)**
-  High serum γ -globulins are observed in: **chronic liver diseases** (hepatitis and cirrhosis), **infection**, and **multiple myeloma**.

3- Bilirubin test:

-  A byproduct of red blood cell breakdown.
-  It is the yellowish pigment observed in jaundice.
-  Normal range of Tb: 0.2 to 1.2 mg/dl
-  Unconjugated/free/indirect (bilirubin-albumin complex):
0.2 to 0.8 mg/dL
-  Conjugated/direct: 0.0 to 0.4 mg/dL
-  High bilirubin levels are observed in: jaundice that mainly resulted from (gallstones, and acute or chronic hepatitis).

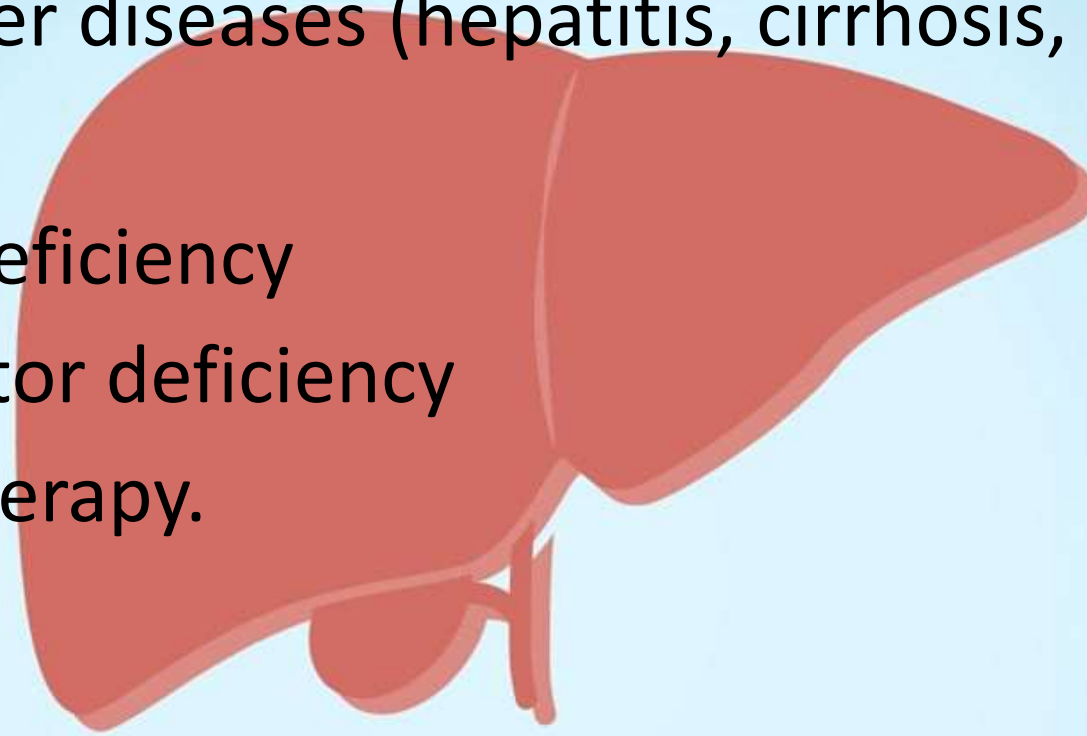
4- Prothrombin time (PT) / International Normalized Ratio (INR):

-  Evaluate the extrinsic pathway of coagulation, which depends on clotting factors (II, V, VII, and X) all of which are made in the liver and required vitamin K.
-  PT: measures how long it takes for a sample of the blood to clot (in seconds).
-  INR: a standardized way of reporting PT (so results comparable across labs).
-  Normal rang: PT (11-13.5) seconds, INR $\approx$ 1.0



Prolonged PT/INR

- found in liver diseases (hepatitis, cirrhosis, acute liver failure).
- vitamin K deficiency
- clotting factor deficiency
- Warfarin therapy.



Clinical Case Reports

Case (1): A healthy 21-year-old student has a routine blood test.

Lab results:

AST: 22 IU/L

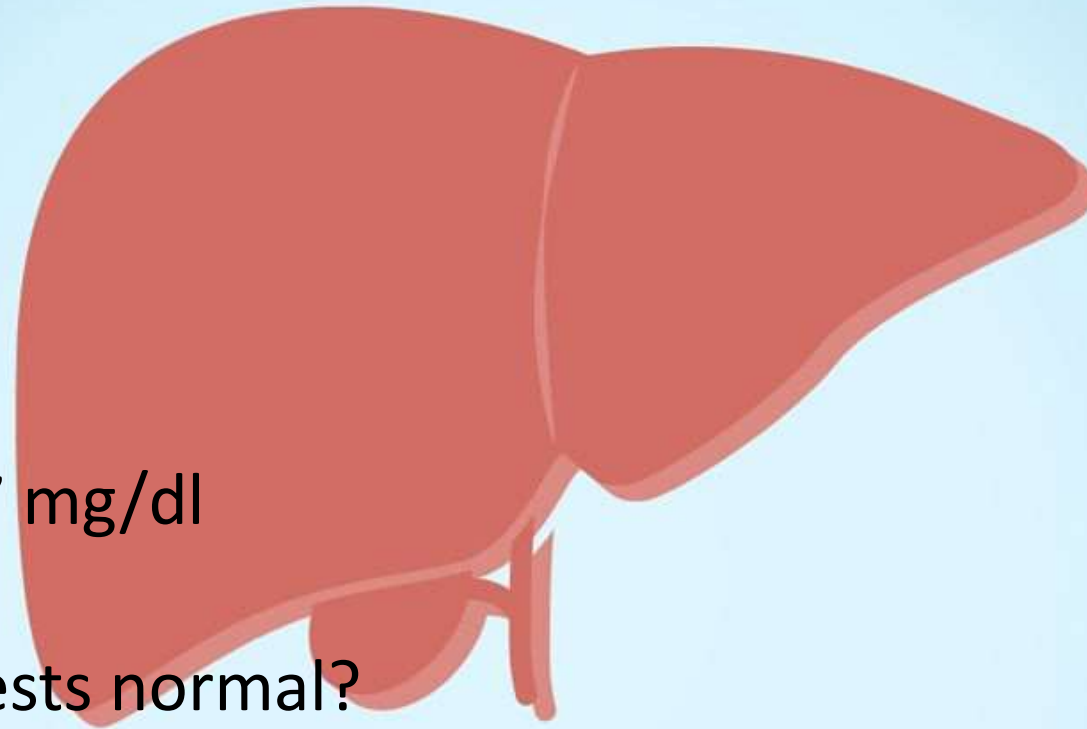
ALT: 25 IU/L

ALP: 90 IU/L

Total bilirubin: 0.7 mg/dl

Q/ Are the liver tests normal?

A/ Yes, all are within the normal range (Normal liver function)



Case (2): A patient has yellow discoloration of the eyes.

Lab results:

AST: 28 IU/L

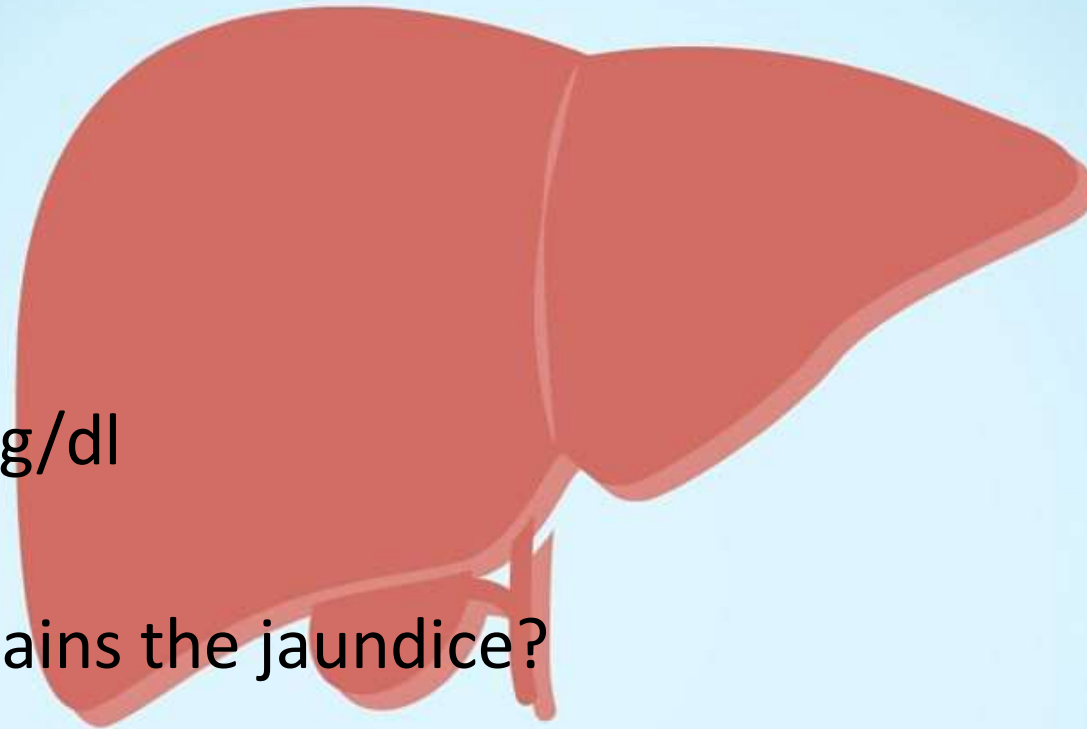
ALT: 30 IU/L

ALP: 95 IU/L

Total bilirubin: 5 mg/dl

Q/ Which test explains the jaundice?

A/ Bilirubin (High bilirubin)



Case (3): A 45-year-old man has abdominal pain and yellow eyes.

Lab results:

AST: 90 IU/L

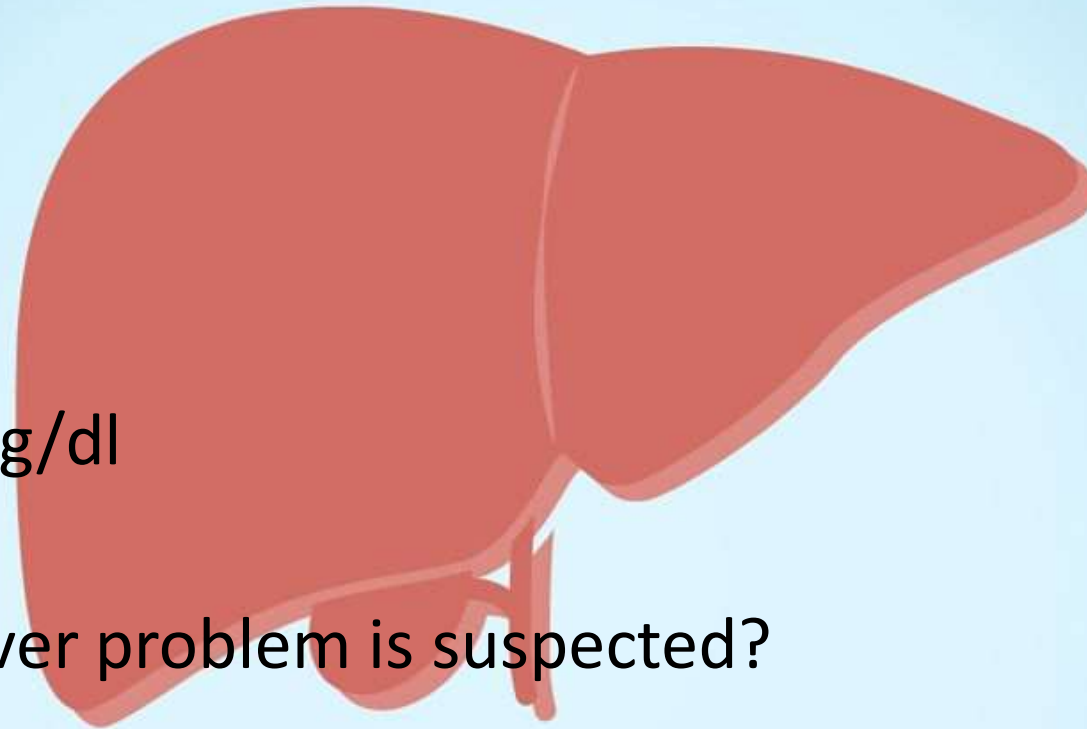
ALT: 100 IU/L

ALP: 500 IU/L

Total bilirubin: 6 mg/dl

Q/ What type of liver problem is suspected?

A/ Cholestasis/ Obstructive jaundice (Gallstone obstruction)



References:

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THANK YOU